# Factory

## Development as a reproducible process

How we turn messy problems into shipped, verified, documented improvements - without losing context.

epics | review loops | verification gates | observability

Starter deck / 2026-05-19

# Why Factory exists

The bottleneck was never imagination. It was execution speed.

Before AI, many valuable engineering tasks were technically possible but economically irrational: writing thousands of tests, exhaustively hardening edge cases, replaying failures, documenting every pattern, and reviewing every boring change with full attention.

### Compress the loop

Move from idea to patch to verification in minutes, not days.

### Keep control

Use tests, specs, gates, traces, and artifacts to constrain AI-generated work.

### Scale the method

Capture the process so more people can operate the same factory.

**Factory = a way to turn AI speed into reliable software delivery**

# What Factory means by reproducible

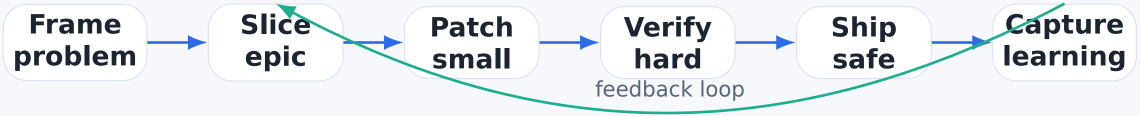
Factory does not claim that every software outcome is bit-for-bit reproducible.

It aims to make the **development process** reproducible:

- how we frame ambiguous problems;
- how we slice work into epics;
- how we preserve context and decisions;
- how we verify changes;
- how we capture reviewer feedback;
- how we turn incidents and fixes into reusable knowledge.

**Reproducible process, bounded system, evidence-driven confidence**

# The process in one sentence




## Operating principle

Every change should leave behind enough evidence for a future engineer to understand what changed, why, how it was verified, and what remains risky.

## Anti-goal

Do not turn development into ceremony. Factory documents the shortest reliable path from ambiguity to confidence.

# Factory converts tacit work into explicit assets

Chat-based investigation
↓
Implementation summary
↓
Reviewer feedback + fixes
↓
Verification transcript
↓
Reusable Factory pattern

**Outputs we want**

- Playbooks for recurring work
- Checklists for risky changes
- Example PRs with narrative
- CI/gate design notes
- Debugging recipes
- Onboarding modules

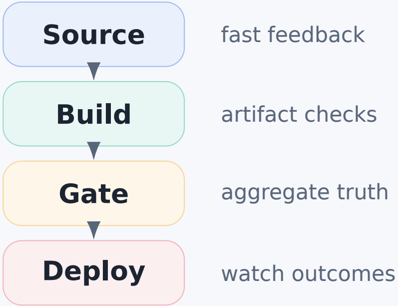
# A Factory epic has a standard shape


| Part | What gets captured |
| --- | --- |
| Problem | Symptom, operator impact, current evidence, constraints. |
| Plan | Small slices, risk boundaries, verification strategy. |
| Closeout | Files changed, tests run, unresolved follow-ups, rollout notes. |

# Verification is part of the product, not a cleanup step

**Factory asks for evidence at each layer**

- Static checks: format, lint, type checks
- Unit and widget tests
- Integration and fixture regressions
- Artifact sanity checks
- Deployment gates and post-deploy observation




**Rule of thumb: if a reviewer can ask “how do we know?”, the Factory artifact should already ans**

# Review loops are structured, not adversarial

## Reviewer role

Find hidden risk, force precision, and turn vague confidence into explicit evidence.

## Implementer role

Respond with scoped patches, verification, and concise summaries.

## Factory role

Preserve the pattern: what kind of issue was caught, how it was fixed, and how it becomes a future checklist item.

## Outcome

Every review improves both the code and the development system.

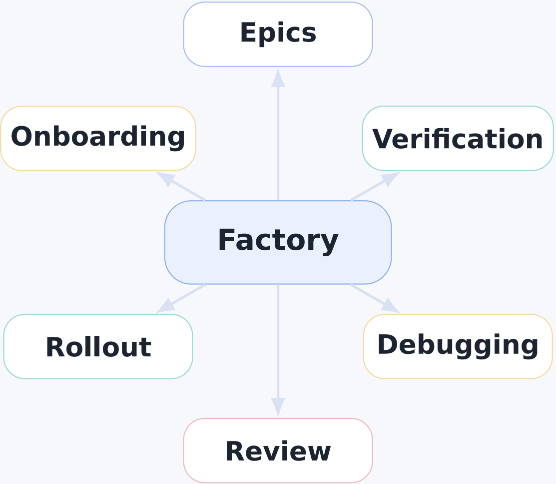
# Instrumentation-first debugging

When a system behaves strangely, Factory prefers visibility over guessing.

**Symptom** → **Hypotheses** → **Instrumentation** → **Trace truth** → **Fix**

Examples: timing breakdowns, trace diagnostics, synthetic/no-trace classifications, debug endpoints, stale artifact flags, deployment gate status files.

# Factory knowledge map



# First content tracks

| Track | Starter artifact | Audience |
|---|---|---|
| Epic lifecycle | How to run an implementation epic from symptom to closeout | Engineers, tech leads |
| Verification gates | Artifact-driven CI gates and deployment safety | Platform, DevOps, reviewers |
| Debugging playbooks | Instrumentation-first trace/debug method | Backend, SRE, support |
| Review patterns | Turning reviewer comments into reusable checks | Everyone shipping code |
| Rollout notes | How to ship risky changes with observability | Leads, operators |

**Start with real examples. Generalize only after the pattern has survived contact with productio**

# Suggested launch plan

### 1. Seed

Create a short deck, a README, and 2-3 concrete case studies from recent work.

### 2. Practice

Use Factory language during active epics: plan, evidence, gate, closeout, follow-up.

### 3. Spread

Turn successful epics into brown bags, onboarding docs, and reusable templates.

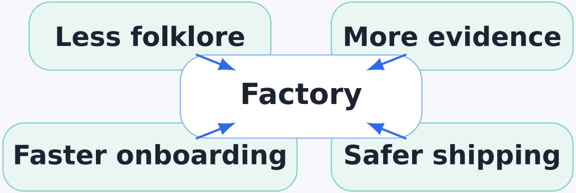
**1**
shared vocabulary

**3**
starter tracks

**30 min**
first internal talk

**ongoing**
living playbook

# What good looks like




The end state is not “more documentation”.
It is a development process that teaches itself.

# FAQ: Factory Asked Questions

### Can you sell me the Factory?

Sell what?
You can build your own.
There is nothing to buy here.
Factory is a process, not a product.

### What do I need to build a Factory?

Feedback loops.
You need to build them, maintain them, and be part of them to some extent.

### What does my Factory need to evolve?

Time and energy.
The process improves only when real work keeps flowing through it.

**Factory is not installed. Factory is grown.**

# Factory

Make the way we build more inspectable, repeatable, and reliable.

**next: turn this deck into the Factory README and first playbook**